

Optimal Racing Line, Energy Management, and Race Strategy for Electric Vehicles under Uncertain Grip

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Abstract

Autonomous racing pushes the limits of control and optimization, requiring vehicles to navigate circuits at the edge of their physical capabilities. For electric vehicles, performance is further constrained by battery energy, making the racing line problem inherently multiobjective, let alone tire grip uncertainty during the race paving the way for a robust stochastic optimization. We formulate the joint optimization of racing line and speed profile on closed circuits and solve it through five stages. First, we compute the time-optimal trajectory using a custom Sequential Convex Programming (SCP) algorithm with a hand-coded Simplex solver, validated against `scipy`'s Sequential Least Squares Programming (SLSQP). Second, we trace the Pareto front between lap time and energy consumption via a weighted-sum formulation. Third, we extend the optimization to uncertain tire grip using Sample Average Approximation (SAA). Then to push beyond this, we model a mass-coupled system taking into account the mass of the battery with (4) battery sizing for multi-lap races under mass-energy coupling, and (5) race strategy co-optimization (including energy, time, and battery sizing) with a closed-form Karush–Kuhn–Tucker (KKT) solution for line choice and pace management. A final layer of optimization is performed implementing a TV-LQR controller for trajectory tracking on the optimal trajectory to ensure dynamic feasibility. We test our pipeline on three different race circuits. Our SCP solution converges in $12\times$ fewer iterations than SLSQP while reaching a comparable solution. Grip uncertainty imposes a $\sim 10\%$ lap-time penalty at the time-optimal operating point to produce a control that is robust through sliding and grip uncertainty, especially because grip uncertainty also strongly diminishes energy efficiency. Pace management reduces the minimum feasible battery capacity by 32% through a closed-form KKT strategy under the same race strategy across multi-lap races.

1 Introduction

Finding the fastest path around a race track is a classical problem in vehicle dynamics and motorsport engineering [1, 2]. The trajectory followed by a vehicle directly determines the curvature encountered along the track and therefore the maximum speed that can be maintained through each corner. By exploiting the full width of the track, drivers reduce curvature and increase cornering speed, leading to significant reductions in lap time.

For electric vehicles, racing line optimization becomes a multi-objective problem. In addition to minimizing lap time, the vehicle must manage a finite energy budget. Aggressive acceleration and high cornering speeds improve performance but increase energy consumption, while conservative driving strategies preserve battery charge at the cost of slower lap times. This tradeoff is particularly relevant in modern electric racing series such as Formula E [3] and in autonomous racing applications.

This project investigates the computation of optimal racing trajectories for electric vehicles under both deterministic and uncertain operating conditions. We formulate the problem as a nonlinear optimization over the racing line and speed profile and study the tradeoff between lap time and energy consumption. We further analyze the impact of uncertain tire grip through a robust optimization framework.

Our approach is organized into five stages: (1) time-optimal racing line optimization using a kinematic vehicle model and Sequential Convex Programming (SCP), (2) multi-objective optimization of lap time and energy consumption through Pareto front analysis, and (3) robust optimization under uncertain tire-road friction using Sample Average Approximation (SAA). Stages 1–3 use a fixed-mass kinematic model for

trajectory optimization. Then for more realistic vehicle dynamics, we model a mass-varying battery as it acts on the total energy available to the vehicle but also increases energy consumption during the race. Hence the two last stages are (4) battery sizing with mass-energy coupling, and (5) closed-form KKT race strategy. All circuits used in this study are described in Appendix F. The optimized trajectories are subsequently tracked using a dynamic bicycle model and a Time-Varying Linear Quadratic Regulator (TV-LQR) to evaluate their feasibility in closed-loop simulation.

2 Problem Formulation

2.1 Optimization Overview

The objective is to jointly optimize the geometric racing line and the corresponding speed profile around a closed circuit. We first compute the minimum-time trajectory considering only kinematic constraints, then extend the formulation to include battery energy consumption and generate a Pareto front, and finally account for uncertainty in tire grip. Figure 1 illustrates the track parameterization and the optimization variables.

2.2 Decision Variables

The racing line is parameterized using lateral offsets $\alpha = (\alpha_1, \dots, \alpha_n)$ measured from the track centerline. At station i , the vehicle position is $\mathbf{p}_i = \mathbf{c}_i + \alpha_i \hat{\mathbf{n}}_i$, where $\mathbf{c}_i$ denotes the centerline position and $\hat{\mathbf{n}}_i$ the outward normal direction. The constraint $|\alpha_i| \leq w_i$ keeps the path within the track of half-width w_i . The speed profile is $\mathbf{v} = (v_1, \dots, v_n)$, and the complete optimization vector is $\mathbf{x} = (\alpha, \mathbf{v}) \in \mathbb{R}^{2n}$.

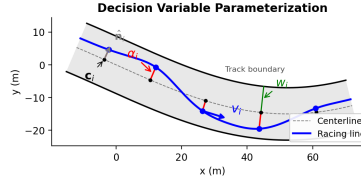


Figure 1: Track parameterization and optimization variables. The lateral offset α_i displaces each path point from the centerline $\mathbf{c}_i$ along the normal $\hat{\mathbf{n}}_i$. The speed v_i is along the tangent; w_i bounds the feasible region.

2.3 Vehicle and Track Dynamics

The optimization uses a kinematic point-mass model that captures the dominant physical limitations. At each station, the centripetal acceleration required to follow the curved path must not exceed the available tire friction:

$$v_i^2 |\kappa_i(\alpha)| \leq 0.85 \mu g \quad (1)$$

where κ_i is the path curvature at station i and 0.85 is a safety margin. Between consecutive stations, longitudinal dynamics are bounded by the vehicle's acceleration and braking capabilities:

$$v_{i+1}^2 \leq v_i^2 + 2 a_{\text{lon}} \Delta s_i, \quad v_i^2 \leq v_{i+1}^2 + 2 a_{\text{brk}} \Delta s_i \quad (2)$$

where $\Delta s_i = \|\mathbf{p}_{i+1} - \mathbf{p}_i\|$ is the segment arc length.

2.4 Lap Time and Energy Objectives

The total lap time is

$$T(\alpha, \mathbf{v}) = \sum_{i=1}^n \frac{\Delta s_i(\alpha)}{v_i} \quad (3)$$

This objective is nonlinear in both α (through Δs_i) and $\mathbf{v}$. At each segment, the traction force accounts for inertial, aerodynamic, and rolling resistance contributions: $F_i = m a_i + \frac{1}{2} \rho C_d A v_i^2 + C_r m g$. The mechanical power is $P_i = F_i \cdot v_i$. Depending on the sign of P_i , the electrical energy integrates P_i/η during drive phases or recovers $|P_i| \cdot \eta_{\text{regen}}$ during braking. Summing over all segments yields the total energy expenditure $E(\alpha, \mathbf{v})$. All notation and vehicle parameters are gathered in Appendix A.

3 Time-Optimal Racing Line Using SCP

The first stage finds the time-optimal racing line and speed profile under constraints (1)–(2). Since both the objective and constraints are nonlinear, we use Sequential Convex Programming (SCP) following the trust-region approach in [4]. At each iteration, we linearize all nonlinear constraints around the current iterate and solve the resulting linear program (LP):

$$\min_{\Delta \mathbf{x}} \mathbf{c}^\top \Delta \mathbf{x} \quad \text{s.t.} \quad \mathbf{A} \Delta \mathbf{x} \leq \mathbf{b}, \quad \boldsymbol{\ell} \leq \Delta \mathbf{x} \leq \mathbf{u} \quad (4)$$

where $\mathbf{c}$ is the linearized objective gradient. The constraint matrix $\mathbf{A} \in \mathbb{R}^{3n \times 2n}$ stacks the Jacobians of the cornering, acceleration, and braking constraints, and $\mathbf{b}$ contains the corresponding right-hand sides at the current iterate (see Appendix C for full expressions). The bounds $(\boldsymbol{\ell}, \mathbf{u})$ enforce both the trust region $|\Delta \alpha_i| \leq \rho$ and the variable limits. We solve this LP with a two-phase Simplex algorithm implemented from scratch following [4]; a custom solver gives direct control over the trust-region stopping criterion and avoids the overhead of general-purpose LP interfaces. The curvature Jacobian $\partial \kappa_i / \partial \alpha_j$ is computed with JAX automatic differentiation. With $n = 80$ stations, which denote the number of discretization points chosen for the decision variables, the LP sub-problem involves $2n = 160$ decision variables and $3n$ linear constraints, keeping each iteration computationally lightweight compared to a full nonlinear solve.

The trust region radius ρ adapts at each iteration, halved when the step causes large constraint violations and grown by a factor of 1.2 when the step improves the objective. To validate the SCP solution, we solve the same problem with scipy’s SLSQP. On the Monza-style circuit (Appendix F, 80 stations, 1115 m), SCP converges from a curvature-minimizing warm start at 44.5 s to 35.9 s in 15 iterations, while SLSQP requires approximately 180 iterations to reach 35.4 s, consistent with both solvers converging toward a qualitatively similar optimum.

4 Time-Energy Pareto Optimization

The second stage addresses the tradeoff between lap time and energy consumption. For an electric vehicle that must complete multiple laps, aggressive time-optimal trajectories can discharge the battery rapidly and may not represent the best overall race strategy. To explore this tradeoff, we use the weighted-sum method [4]:

$$\min_{\alpha, \mathbf{v}} (1 - w) \frac{T}{T_{\text{ref}}} + w \frac{E}{E_{\text{ref}}} \quad (5)$$

where T_{ref} and E_{ref} are reference values from the time-optimal solution. Sweeping $w \in [0, 0.95]$ traces the Pareto front.

When we attempted to compute the Pareto front with SCP, the algorithm failed for most $w > 0$. As the energy weight increases, the optimal speed profile shifts far from the warm start. The linearized constraints become poor approximations, causing violations that trigger trust region shrinkage until convergence stalls. We therefore use scipy’s SLSQP for the Pareto computation. The resulting Pareto front is shown in Figure 2b.

5 Robust Optimization under Uncertain Grip

In practice, tire grip varies along the track due to wet patches, rubber deposits, or temperature gradients. We model this by treating the friction coefficient as a per-station random variable: $\mu_i \sim \mathcal{N}(1.0, \sigma_\mu^2)$, clipped to $[0.3, 1.5]$, with $\sigma_\mu = 0.1$.

We handle the stochastic constraints with a sampled worst-case robustification [4]: we draw $N = 20$ grip scenarios $\boldsymbol{\mu}^{(k)}$ and, exploiting the monotonicity of the cornering constraint in μ , tighten each constraint to the sampled worst-case grip at each station:

$$v_i^2 |\kappa_i| \leq \mu_i^{\text{worst}} \cdot g \cdot 0.85, \quad \mu_i^{\text{worst}} = \min_k \mu_i^{(k)} \quad (6)$$

This reduces $N \times n$ scenario constraints to n deterministic constraints with modified grip, making the robust problem no more expensive than the nominal one.

6 Battery Sizing for Multi-Lap Racing

For a full race, the battery must supply enough energy to complete all laps above $\text{SoC}_{\min} = 5\%$. Since pack weight enters vehicle mass directly ($m = m_{\text{chassis}} + Q_{\text{batt}}/e_{\text{spec}}$), a larger battery both increases available energy and raises per-lap consumption, coupling the sizing and performance problems. We sweep $Q_{\text{batt}} \in [15, 80]$ kWh; at each point, we simulate the lap energy E_{lap} from the kinematic speed profile using a parabolic motor efficiency map (peak $\eta = 0.95$ at 60% rated power). A pack is feasible when

$$Q_{\text{batt}} (1 - \text{SoC}_{\min}) \geq n_{\text{laps}} \cdot E_{\text{lap}}(Q_{\text{batt}}) \quad (7)$$

and the minimum feasible capacity $Q^* = \min\{Q_{\text{batt}} : \text{feasible}\}$ is the design target.

7 Race Strategy Co-Optimization

We introduce two per-lap decision variables: a Pareto operating point $g \in [g_{\min}, g_{\max}]$ controlling cornering aggressiveness, and a pace factor $p \in [p_{\min}, 1]$ scaling speed uniformly. Lap time and energy scale as $T_{\text{lap}} = T_{\text{base}}(g)/p$ and $E_{\text{lap}} = E_{\text{base}}(g) p$, subject to the global energy budget:

$$\sum_{t=1}^n E_{\text{base}}(g_t) p_t \leq Q_{\text{batt}} (1 - \text{SoC}_{\min}) \quad (8)$$

For a symmetric race, KKT stationarity and complementary slackness [4] lead to a closed-form structure of the optimum. The energy constraint is active at optimality, and the optimal racing line minimizes the coupled time–energy metric:

$$g^* = \underset{g}{\operatorname{argmin}} T_{\text{base}}(g) E_{\text{base}}(g), \quad p^* = \frac{Q_{\text{batt}} (1 - \text{SoC}_{\min})}{n \cdot E_{\text{base}}(g^*)} \quad (9)$$

with p^* clipped to $[p_{\min}, 1]$ and subject to the competitiveness constraint $T_{\text{base}}(g) \leq 1.15 T_{\min}$. An NLP validation (SLSQP) confirms a solution consistent with the constant (g^*, p^*) structure predicted by the KKT conditions, supporting the closed-form result under symmetric-race assumptions.

This derivation relies on three key assumptions: (i) all laps are identical (symmetry); (ii) linear scaling laws $T_{\text{lap}} = T_{\text{base}}/p$ and $E_{\text{lap}} = E_{\text{base}} p$, which are exact under uniform speed scaling but approximate when aerodynamic drag dominates; and (iii) the function $T_{\text{base}}(g) E_{\text{base}}(g)$ is unimodal over the feasible range, as consistently observed in numerical experiments.

8 Dynamic Trajectory Tracking

The kinematic optimization produces a racing line and speed profile, but these are not directly executable by a vehicle with tire slip and yaw dynamics. To bridge this gap, we formulate the trajectory tracking as a finite-horizon optimal control problem (full dynamics in Appendix B).

Given a reference trajectory $(\bar{\mathbf{s}}_k, \bar{\mathbf{u}}_k)_{k=0}^N$, we seek the control sequence that minimizes tracking error:

$$\min_{\mathbf{u}_{0:N-1}} \sum_{k=0}^{N-1} [(\mathbf{s}_k - \bar{\mathbf{s}}_k)^\top Q (\mathbf{s}_k - \bar{\mathbf{s}}_k) + (\mathbf{u}_k - \bar{\mathbf{u}}_k)^\top R (\mathbf{u}_k - \bar{\mathbf{u}}_k)] \quad (10)$$

subject to the bicycle dynamics $\mathbf{s}_{k+1} = f(\mathbf{s}_k, \mathbf{u}_k)$, where $\mathbf{s} = [x, y, \psi, v_x, v_y, \omega]^\top$ and $\mathbf{u} = [\delta, F_{\text{drive}}]^\top$.

We solve this in two phases. First, a Stanley lateral controller tracks the kinematic racing line to generate a dynamically feasible reference $(\bar{\mathbf{s}}_k, \bar{\mathbf{u}}_k)$ consistent with the bicycle model. Second, we linearize the dynamics around each reference point via finite differences to obtain $A_k = \partial f / \partial \mathbf{s}$ and $B_k = \partial f / \partial \mathbf{u}$, then solve the discrete Riccati equation backward to compute the TV-LQR gains K_k (see Appendices D and E for the full algorithm). Online, the control law is

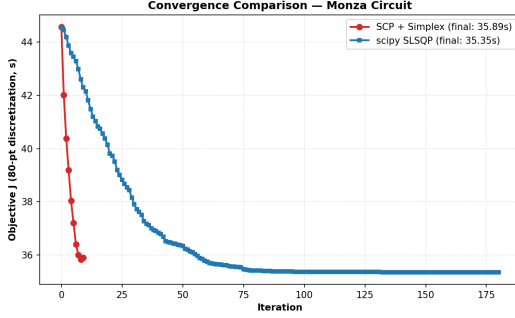
$$\mathbf{u}_k = \bar{\mathbf{u}}_k + K_k (\mathbf{s}_k - \bar{\mathbf{s}}_k) \quad (11)$$

with spatial indexing to handle timing mismatches between the reference and the actual vehicle state.

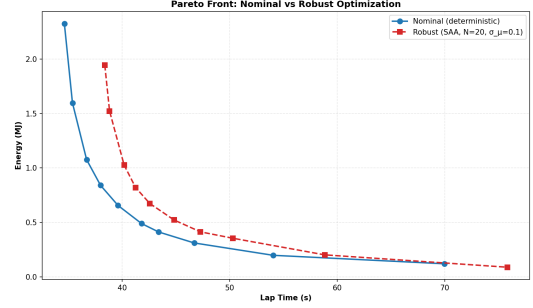
9 Results

9.1 SCP vs SLSQP Convergence

Figure 2a shows the convergence of both solvers on the Monza-style circuit. Starting from a curvature-minimizing warm start at 44.5 s, SCP reaches 35.9 s in 15 iterations while SLSQP requires approximately 180 iterations to reach 35.4 s. The ~ 0.5 s gap is the expected cost of SCP’s conservative trust region: it trades a small residual suboptimality for a $12\times$ reduction in iterations.



(a) Convergence on the Monza-style circuit (80 stations, 1115 m).



(b) Nominal and robust Pareto fronts (Monza-style circuit, Appendix F).

Figure 2: (a) SCP converges substantially faster than SLSQP while reaching a qualitatively similar solution. (b) The robust Pareto front is shifted to longer lap times due to worst-case grip constraints; the gap narrows for energy-efficient operating points.

9.2 Time-Energy Pareto Front

Figure 2b presents the nominal Pareto front, which derives from the optimization problem without uncertainty, and the robust Pareto front, which derives from the stochastic optimization problem, both on the Monza-style circuit (Appendix F). At $w = 0$, the nominal optimum reaches ~ 32 s (consistent with an independent SCP run on this circuit). As w increases, the optimizer trades lap time for energy savings, reaching ~ 70 s at $w = 0.95$ with near-zero net energy. The robust front is shifted to longer lap times since at $w = 0$, the robust optimum is ~ 36 s, a 4 s penalty that quantifies the cost of robustness. However, this penalty shrinks as energy becomes more important in the objective, since energy-efficient trajectories are inherently smoother with lower cornering loads, reducing the sensitivity to grip uncertainty.

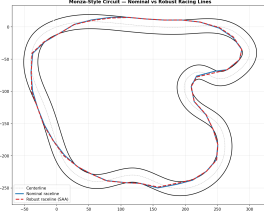
9.3 Nominal vs Robust Trajectories

Figure 3 compares the nominal and robust racing lines and speed profiles on the Monza-style circuit (Appendix F). The geometric racing lines are nearly identical since curvature reduction is grip-independent. The robust speed profile, however, is consistently lower, reflecting the conservatism of the sampled worst-case grip constraint. Yet the profile retains the same shape and variations as the nominal one since the vehicle still accelerates on straights and brakes into corners, only at reduced magnitudes.

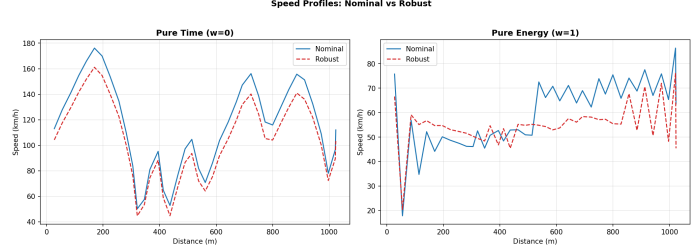
9.4 Battery Sizing

Figure 4 presents the battery sizing sweep for the Grand Prix circuit (Appendix F). The left panel shows the energy balance as a function of battery capacity Q_{batt} . Below a critical threshold $Q^* \approx 37$ kWh, the required energy exceeds the available capacity and the mission becomes infeasible. Above this limit, a growing safety margin appears, with a final state of charge remaining above the minimum constraint (5%).

At the feasibility boundary, the vehicle operates at the limit of its energy budget, with a mass of approximately 824 kg and a final state of charge around 7%. Slight undersizing leads to mission failure, while oversizing introduces increasing dead weight with limited performance benefit. This sharp transition highlights a strongly nonlinear feasibility boundary driven by the coupling between mass and energy consumption.



(a) Racing lines.



(b) Speed profiles.

Figure 3: Nominal vs robust trajectories. The geometric paths are similar, but the robust profile maintains lower speeds throughout.

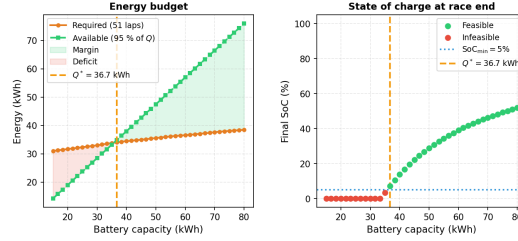


Figure 4: Battery sizing sweep. Left: required vs. available energy as a function of battery capacity, identifying the feasibility threshold $Q^* \approx 37$ kWh. Right: final state of charge at race end, showing a sharp transition between infeasible ($\text{SoC} \leq 0$) and feasible regimes.

9.5 Race Strategy, Battery Reduction, and Circuit Comparison

Figure 5 summarizes the joint optimization of race strategy and circuit characteristics. The left panel shows the time–energy Pareto front obtained by varying the grip fraction $g \in [0.50, 0.90]$. The fastest configuration ($g = 0.90$) maximizes energy consumption (677 Wh/lap), while moderate pace relaxation within a $\sim 15\%$ time penalty yields up to $\sim 8\%$ reduction in energy use.

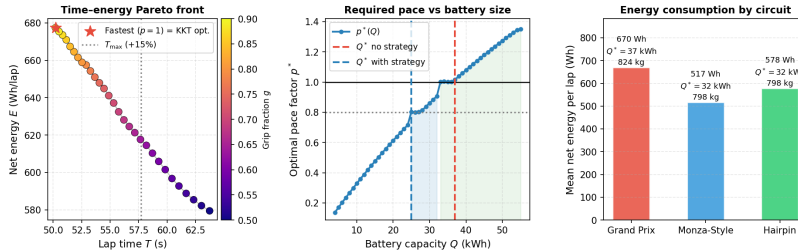


Figure 5: Race strategy and circuit comparison. Left: time–energy Pareto front. Middle: optimal pace law showing how strategic pacing reduces the required battery capacity. Right: comparison of circuits in terms of mean energy per lap and resulting battery sizing requirements.

The middle panel illustrates the optimal pacing strategy as a function of battery capacity. Allowing pace modulation reduces the minimum feasible battery size from 37 to 25 kWh, corresponding to a 32% reduction and approximately 60 kg of mass savings. The solution structure matches the constant (g^*, p^*) regime predicted by the KKT conditions, confirming the optimality of the numerical solution.

The right panel compares different circuit layouts over a fixed ~ 65 km race distance. The Grand Prix circuit requires approximately 670 Wh/lap and a battery size of 37 kWh (total weight of 824 kg). The Monza-style circuit is the most efficient at 517 Wh/lap, reducing the required battery to 32 kWh (798 kg), a 23% improvement relative to the baseline. The Hairpin circuit lies in between at 578 Wh/lap and 32 kWh. Overall, circuit geometry is a dominant driver of energy demand and therefore battery sizing.

10 Conclusion

This work presented a five-stage hierarchical optimization framework for electric vehicle racing. The SCP formulation efficiently solved the minimum-time problem while substantially reducing computational

effort compared with a standard nonlinear optimization approach. Multi-objective optimization revealed clear trade-offs between lap time, energy consumption, and robustness to grip uncertainty, showing that modest increases in lap time can enable significant energy savings while reducing sensitivity to adverse conditions.

Battery sizing and race strategy optimization further demonstrated that jointly optimizing racing line aggressiveness and pace can substantially reduce energy requirements. In the Grand Prix case (Appendix F) considered, co-optimization reduced the required battery capacity by about 32%, corresponding to roughly 60 kg of mass savings, while maintaining competitive performance. Across the circuits considered, optimal battery size remained strongly influenced by track geometry and curvature.

A key limitation of the present framework lies in the separation between kinematic optimality, where we consider cars as point masses, and dynamic feasibility, where we use a bicycle model with yaw dynamics. Hence the pre-computed optimal times and energy budgets from the kinematic model are approximate, and the actual closed-loop performance depends on the controller’s ability to track the reference trajectory.

To balance fidelity and tractability, a two-level modeling hierarchy was adopted with a kinematic formulation for trajectory optimization and a mass-coupled model for battery sizing and strategy optimization. This decomposition enables efficient computation while preserving the dominant physical trade-offs.

Future work could focus on integrating these modeling levels into a unified formulation, reducing conservatism through chance-constrained approaches, and relaxing the symmetric-race assumption in the KKT-based strategy formulation since the car does not necessarily follow the exact same trajectory at every lap in real-world races. Additional race effects such as tire degradation or race interruptions could also be incorporated, including pit stops for tire changes, battery thermal management, or adapting to changing weather conditions.

Contributions

Erwin Poussi developed the simulator, the SCP algorithm with Simplex and the scipy SLSQP pipeline, the single-lap time-energy Pareto front computation and sweep strategy, the race simulation environment (rendering, vehicle dynamics, battery discharge physics and discretization), the stochastic optimization formulation, and the TV-LQR trajectory tracking controller. Matthieu Hautsch implemented the multi-lap time-energy Pareto front analysis, the battery sizing sweep, and the race strategy co-optimization including the KKT analytical solution, the NLP validation, and the multi-circuit comparison. Code available at <https://github.com/Rwin2/EV-optimal-line-racing>.

References

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Supplementary Material

A Notation and Vehicle Parameters

Symbol	Value	Description
m	1600 kg	Vehicle mass (Stages 1–3)
L	2.9 m	Wheelbase
l_f, l_r	1.4, 1.5 m	CG to front/rear axle
I_z	2500 kg m ²	Yaw moment of inertia
μ	1.0	Tire-road friction coefficient
C_f, C_r	80 000, 90 000 N/rad	Cornering stiffness (front/rear)
C_d	0.30	Aerodynamic drag coefficient
A	1.5 m ²	Frontal area
ρ	1.225 kg/m ³	Air density
c_r	0.012	Rolling resistance coefficient
$P_{\max}$	250 kW	Maximum motor power
$F_{\text{drive},\max}$	8 000 N	Maximum traction force
$F_{\text{brake},\max}$	12 000 N	Maximum braking force
η	0.92	Motor efficiency
η_{regen}	0.65	Regenerative braking efficiency
g	9.81 m/s ²	Gravitational acceleration

Table 1: Vehicle parameters for Stages 1–3 (SCP, Pareto, robust optimization).

Symbol	Value	Description
m_{chassis}	640 kg	Chassis + motor + electronics mass
e_{spec}	200 Wh/kg	Battery system-level specific energy (Formula E Gen3)
$P_{\max}$	300 kW	Maximum motor power
$F_{\text{drive},\max}$	10 000 N	Maximum traction force
$F_{\text{brake},\max}$	15 000 N	Maximum braking force
η_{peak}	0.95	Peak motor efficiency
η_{regen}	0.65	Regenerative braking efficiency
$\text{SoC}_{\min}$	0.05	Minimum allowed state of charge
$p_{\min}$	0.80	Minimum pace factor (strategy)

Table 2: Vehicle parameters for Stages 4–5 (battery sizing and race strategy), based on Formula E Gen3 regulations.

B Bicycle Model Dynamics

The dynamic bicycle model used for trajectory tracking has state $\mathbf{s} = [x, y, \psi, v_x, v_y, \omega]^\top$ and control $\mathbf{u} = [\delta, F_{\text{drive}}]^\top$. The continuous-time equations of motion are:

$$\dot{x} = v_x \cos \psi - v_y \sin \psi \quad (12)$$

$$\dot{y} = v_x \sin \psi + v_y \cos \psi \quad (13)$$

$$\dot{\psi} = \omega \quad (14)$$

$$\dot{v}_x = \frac{1}{m} (F_{\text{drive}} - F_{\text{drag}} - F_{\text{roll}} + m v_y \omega - F_{yf} \sin \delta) \quad (15)$$

$$\dot{v}_y = \frac{1}{m} (F_{yf} \cos \delta + F_{yr} - m v_x \omega) \quad (16)$$

$$\dot{\omega} = \frac{1}{I_z} (l_f F_{yf} \cos \delta - l_r F_{yr}) \quad (17)$$

where the lateral tire forces use a linear slip angle model with saturation:

$$\alpha_f = \delta - \arctan \frac{v_y + l_f \omega}{v_x}, \quad \alpha_r = -\arctan \frac{v_y - l_r \omega}{v_x} \quad (18)$$

$$F_{yf} = \text{clip}(C_f \alpha_f, -F_{\text{lim}}, F_{\text{lim}}), \quad F_{yr} = \text{clip}(C_r \alpha_r, -F_{\text{lim}}, F_{\text{lim}}) \quad (19)$$

with $F_{\text{lim}} = \frac{1}{2} \mu m g$. The resistive forces are $F_{\text{drag}} = \frac{1}{2} \rho C_d A v_x^2$ and $F_{\text{roll}} = c_r m g$. The drive force is further limited by power: $F_{\text{drive}} \leq P_{\text{max}}/v_x$ when $F_{\text{drive}} > 0$. These equations are integrated using a fourth-order Runge-Kutta scheme at $\Delta t = 0.02$ s.

C SCP Sub-Problem Construction

At SCP iteration with current iterate $(\alpha, \mathbf{v})$, the LP (4) is constructed as follows. Let $\Delta \mathbf{x} = (\Delta \alpha, \Delta \mathbf{v})$.

Objective gradient. The linearized lap time gradient is:

$$\mathbf{c} = \begin{bmatrix} \frac{\partial \mathbf{ds}^\top}{\partial \alpha} \frac{1}{\mathbf{v}} \\ \mathbf{ds} \\ -\frac{1}{\mathbf{v}^2} \end{bmatrix} \quad (20)$$

where $\mathbf{ds}$ is the vector of segment lengths and the Jacobian $\partial \mathbf{ds} / \partial \alpha$ is computed from the path geometry.

Cornering constraints. Linearizing $v_i^2 |\kappa_i| \leq a_{\text{lat}}$ around the current iterate yields rows of $\mathbf{A}$ involving $v_i^2 \cdot \text{sgn}(\kappa_i) \cdot \partial \kappa_i / \partial \alpha_j$ (from JAX autodiff) and $2 v_i |\kappa_i|$.

Acceleration/braking constraints. Similarly linearized from (2), coupling consecutive speed variables through the segment-length Jacobian.

D TV-LQR Gain Computation

Given the reference trajectory $(\bar{\mathbf{s}}_k, \bar{\mathbf{u}}_k)$, we compute the discrete Jacobians $A_k \in \mathbb{R}^{6 \times 6}$ and $B_k \in \mathbb{R}^{6 \times 2}$ by finite differences of the RK4 integrator. The TV-LQR gains are obtained by solving the discrete Riccati equation backward:

$$K_k = -(R + B_k^\top P_{k+1} B_k)^{-1} B_k^\top P_{k+1} A_k \quad (21)$$

$$P_k = Q + A_k^\top P_{k+1} (A_k + B_k K_k) \quad (22)$$

with terminal condition $P_N = Q$. The weight matrices are $Q = \text{diag}(5, 5, 10, 1, 0.5, 0.5)$ and $R = \text{diag}(10, 10^{-5})$, penalizing position and heading errors more heavily than lateral velocity, while keeping steering effort smooth.

E Stanley + TV-LQR Controller Algorithm

The online controller combines a Stanley lateral controller with TV-LQR feedback gains precomputed offline. Algorithm 6 summarizes the procedure.

F Test Circuits

All circuits are procedurally generated closed tracks; lap counts are chosen to normalize race distance to ≈ 65 km. Figure 7 shows their layouts.

1. Offline (reference generation):

- Simulate the bicycle model tracking the optimized raceline with a Stanley controller to produce a dynamically feasible reference $(\bar{\mathbf{s}}_k, \bar{\mathbf{u}}_k)$.
- Linearize dynamics at each reference point via finite differences of the RK4 integrator to obtain A_k, B_k .
- Solve the discrete Riccati equation backward (21)–(22) to obtain TV-LQR gains K_k .

2. Online (at each time step):

- Find the nearest reference point using spatial indexing (closest point on the raceline).
- Compute the Stanley steering command δ_{stanley} from cross-track error and heading error.
- Compute the LQR correction $\Delta \mathbf{u} = K_k(\mathbf{s} - \bar{\mathbf{s}}_k)$, scaled by a blending factor that decreases with position error.
- Apply the combined control $\delta = \delta_{\text{stanley}} + 0.15 \Delta \delta_{\text{lqr}}$, $F_{\text{drive}} = \bar{F}_k + \Delta F_{\text{lqr}}$.
- Enforce grip margin (95% of friction limit) and power constraints.

Figure 6: Stanley + TV-LQR trajectory tracking algorithm.

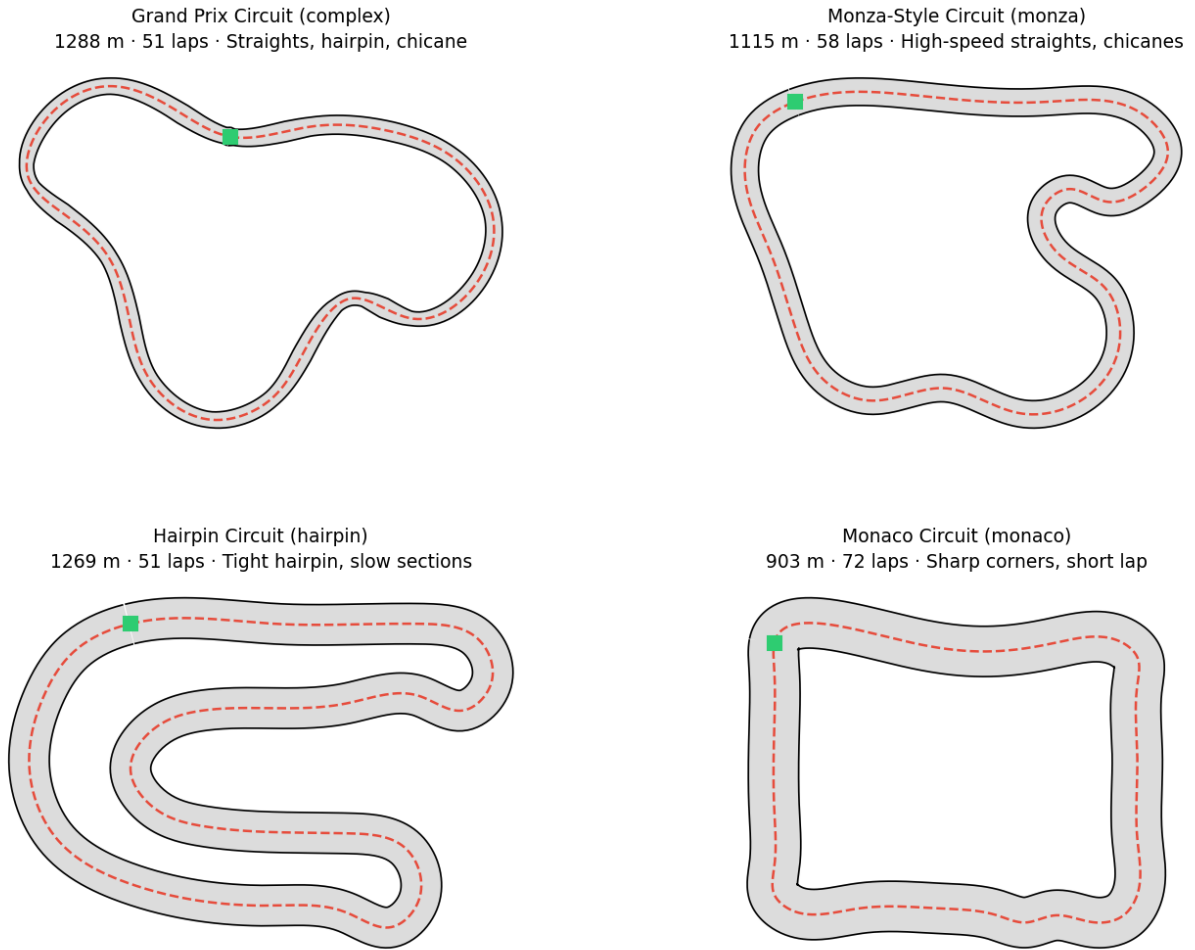


Figure 7: Track layouts. Red dashed line: centerline; green square: start/finish. Each panel lists the circuit name, length, lap count for a ≈ 65 km race, and key characteristics.